

USING DDL STATEMENTS TO CREATE AND MANAGE TABLES

1. Create table DEPT1 with the following columns: DEPTNO INTEGER PRIMARY KEY, DNAME VARCHAR2(30) NOT NULL, LOC VARCHAR2(30) NOT NULL.

SQL> create table dept1(deptno integer primary key,dname varchar2(30) not null, loc varchar2(30) not null);

```
SQL> create table dept1(deptno integer primary key,dname varchar2(30) not null, loc varchar2(30) not null);
Table created.
SQL> desc dept1;
Name                               Null?    Type
-----
DEPTNO                             NOT NULL NUMBER(38)
DNAME                              NOT NULL VARCHAR2(30)
LOC                                NOT NULL VARCHAR2(30)
SQL> |
```

2. Create table EMP1 with the following columns: EMPNO INTEGER PRIMARY KEY, ENAME VARCHAR2(20) NOT NULL, SAL NUMBER(10,2) CHECK (SAL > 5000), MGR NUMBER (Foreign Key referencing EMP1.EMPNO), DEPTNO INTEGER (Foreign Key referencing DEPT1.DEPTNO).

SQL> create table emp1(empno integer primary key,ename varchar2(20) not null,sal number(10,2) check (sal>5000),mgr number(sal>5000),deptno integer, foreign key(deptno) references dept1(deptno));

```
SQL> create table emp1(empno integer primary key,ename varchar2(20) not null,sal number(10,2) check (sal>5000),mgr number,deptno integer, foreign key(deptno) references dept1(deptno));
Table created.
SQL> desc emp1;
Name                               Null?    Type
-----
EMPNO                             NOT NULL NUMBER(38)
ENAME                              NOT NULL VARCHAR2(20)
SAL                                NUMBER(10,2)
MGR                                NUMBER
DEPTNO                             NUMBER(38)
```

3. Create tables DEPT11 and EMP11 by copying DEPT1 and EMP1.

SQL> create table dept11 as select * from dept1;

```
SQL> create table dept11 as select * from dept1;
```

Table created.

```
SQL> desc dept11;
```

Name	Null?	Type
DEPTNO		NUMBER(38)
DNAME	NOT NULL	VARCHAR2(30)
LOC	NOT NULL	VARCHAR2(30)

```
SQL> create table emp11 as select * from emp1;
```

Table created.

```
SQL> desc emp11;
```

Name	Null?	Type
EMPNO		NUMBER(38)
ENAME	NOT NULL	VARCHAR2(20)
SAL		NUMBER(10,2)
MGR		NUMBER
DEPTNO		NUMBER(38)

4. Add a new column ADDRESS VARCHAR2(30) to the EMP1 table.

```
SQL> alter table emp1 add Address varchar2(30);
```

```
SQL> alter table emp1 add Address varchar2(30);
```

Table altered.

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
ENAME	NOT NULL	VARCHAR2(20)
SAL		NUMBER(10,2)
MGR		NUMBER
DEPTNO		NUMBER(38)
ADDRESS		VARCHAR2(30)

5. Rename the SAL column in EMP1 to SALARY.

```
SQL> alter table emp1 rename column sal to salary;
```

```
SQL> alter table emp1 rename column sal to salary;
```

Table altered.

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
ENAME	NOT NULL	VARCHAR2(20)
SALARY		NUMBER(10,2)
MGR		NUMBER
DEPTNO		NUMBER(38)
ADDRESS		VARCHAR2(30)

6. Rename the foreign key constraint names in EMP1 and verify the constraint names.

```
SQL> SELECT constraint_name,  
           constraint_type,  
           table_name  
FROM user_constraints  
WHERE table_name = 'EMP1';
```

```
SQL> ALTER TABLE EMP1 RENAME CONSTRAINT SYS_C007845 TO FK_EMP1;
```

Table altered.

7. Modify ENAME Column by increasing the length of the column to 40.

```
SQL> alter table emp1 modify ename varchar2(40);
```

```
SQL> alter table emp1 modify ename varchar2(40);
```

Table altered.

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
ENAME	NOT NULL	VARCHAR2(40)
SALARY		NUMBER(10,2)
MGR		NUMBER
DEPTNO		NUMBER(38)
ADDRESS		VARCHAR2(30)

8. Drop the NOT NULL constraint on the ENAME column.

```
SQL> ALTER TABLE emp1 MODIFY ename varchar2(40) NULL;
```

```
SQL> ALTER TABLE emp1 MODIFY ename varchar2(40) NULL;

Table altered.

SQL> desc emp1;
Name                                         Null?    Type
-----
EMPNO                                         NOT NULL NUMBER(38)
ENAME                                         VARCHAR2(40)
SALARY                                        NUMBER(10,2)
MGR                                           NUMBER
DEPTNO                                        NUMBER(38)
ADDRESS                                       VARCHAR2(30)

SQL>
```

9.Create a comment on the DEPT1 table as 'Depts of WIPRO'.

```
SQL> comment on table dept1 is 'Depts of WIPRO';
```

Comment created.

10.Create a comment on the DEPTNO column of DEPT1 as 'Deptno of WIPRO'. SQL>
COMMENT ON COLUMN dept1.deptno is 'deptno of WIPRO';

Comment created.

11. Create a comment on the EMP1 table as 'Employees of WIPRO'.

```
SQL> comment on table emp1 is 'Employees of WIPRO';
```

Comment created.

12.Create a comment on the EMPNO column of EMP1 as 'Empno of WIPRO'.

```
SQL> COMMENT ON COLUMN emp11.empno is 'empno of WIPRO';
```

Comment created.

13.Remove all comments from the tables and columns.

```
SQL> COMMENT ON table emp1 IS '';
```

```
SQL> COMMENT ON COLUMN emp1.empno IS ' ';
```

```
SQL> COMMENT ON table dept1 IS ' ';
```

```
SQL> COMMENT ON COLUMN dept1.deptno IS ' ';
```

14. Mark the SALARY and ENAME columns of EMP1 as UNUSED.

```
ALTER TABLE emp1 SET UNUSED (salary,ename);
```

```
SQL> ALTER TABLE emp1 SET UNUSED (salary,ename);
```

```
Table altered.
```

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
MGR		NUMBER
DEPTNO		NUMBER(38)
ADDRESS		VARCHAR2(30)

```
SQL>
```

15. Drop the UNUSED columns from EMP1.

```
ALTER TABLE emp1 DROP UNUSED COLUMNS;
```

16. Drop the EMP1 and DEPT1 tables.

```
SQL> drop table emp1;
```

```
Table dropped.
```

```
SQL> drop table dept1;
```

```
Table dropped.
```

```

SQL> drop table emp1;

Table dropped.

SQL> drop table dept1;

Table dropped.

SQL> desc emp1;
ERROR:
ORA-04043: object emp1 does not exist

SQL> desc dept1;
ERROR:
ORA-04043: object dept1 does not exist

```

17. Create table EMP1 from the EMP table by copying both the structure and data. Verify the data.

```
SQL> create table emp1 as select * from emp;
```

Table created.

```

SQL> create table emp1 as select * from emp;

Table created.

SQL> select * from emp1;

```

EMPID	ENAME	JOB	MGR	HIREDATE	SAL	COMM
7839 10	KING	PRESIDENT		17-NOV-81	5000	
7788 30	BLAKE	MANAGER	7839	01-MAY-81	2850	
7799 30	Johnson	MANAGER	7342	01-JUN-87	2855	

18. Rename EMP1 to EMP_TEST.

```
SQL> alter table emp1 rename to emp_test;
```

Table altered.

```
SQL> alter table emp1 rename to emp_test;
```

Table altered.

```
SQL> select * from emp_test;
```

EMPID	ENAME	JOB	MGR	HIREDATE	SAL	COMM
7839 10	KING	PRESIDENT		17-NOV-81	5000	
7788 30	BLAKE	MANAGER	7839	01-MAY-81	2850	
7799 30	Johnson	MANAGER	7342	01-JUN-87	2855	

19. Truncate the EMP_TEST table and verify that all rows have been deleted.

```
SQL> truncate table emp_test;
```

```
SQL> select * from emp_test;
```

```
SQL> truncate table emp_test;
```

Table truncated.

```
SQL> select * from emp_test;
```

no rows selected

20. Create table EMP2 from EMP by copying only the EMPNO, ENAME, and SAL columns along with their data.

```
SQL> create table emp2 as select empid,ename,sal from emp;
```

```
SQL> create table emp2 as select empid,ename,sal from emp;
```

Table created.

```
SQL> desc emp2;
```

Name	Null?	Type
EMPID		NUMBER(4)
ENAME		VARCHAR2(10)
SAL		NUMBER(7,2)

21.Drop the EMP2 table.

```
SQL> drop table emp2;
```

Table dropped.

```
SQL> drop table emp2;
```

Table dropped.

```
SQL> desc emp2;
```

```
ERROR:  
ORA-04043: object emp2 does not exist
```

22.Create the EMP2 table again from EMP without copying any data.

```
SQL> CREATE TABLE emp2 AS SELECT * FROM emp WHERE 1=0;
```

Table created.

```
SQL> desc emp2;
```

Name	Null?	Type
EMPID		NUMBER(4)
ENAME		VARCHAR2(10)
JOB		VARCHAR2(9)
MGR		NUMBER(4)
HIREDATE		DATE
SAL		NUMBER(7,2)
COMM		NUMBER(7,2)

DEPTID

NUMBER(2)

SQL> select * from emp2;

no rows selected

23.Drop the EMP2 table.

SQL> drop table emp2;

Table dropped.

24.Flashback the first dropped EMP2 table (containing data) and verify the table.

SQL> FLASHBACK TABLE emp2 TO BEFORE DROP;

```
SQL> SHOW RECYCLEBIN;
ORIGINAL NAME      RECYCLEBIN NAME      OBJECT TYPE  DROP TIME
-----
DEPT               BIN$pTu+QthxRDurrePEUZZTWA==$0  TABLE      2024-09-12:10:21:56
DEPT1              BIN$YvNpjWiZS4Gu3dykRlS+/Q==$0  TABLE      2024-09-28:14:25:04
DEPT1              BIN$Aj02V40eRC+/rPLFTKIaoQ==$0  TABLE      2024-09-21:11:49:11
DEPT11             BIN$17Wv73TcRZWYb0rhj85+1A==$0  TABLE      2024-09-21:11:59:53
EMP                BIN$oJI9pKsFQC6EOdPMIvWrLA==$0  TABLE      2024-09-12:10:21:49
EMP1               BIN$Dxs3bh7vSo0Uvq+EmjQ1Nw==$0  TABLE      2024-09-28:14:31:22
EMP1               BIN$b5ai/y0bQ5ScvsrR7oVkoQ==$0  TABLE      2024-09-28:14:24:57
EMP2               BIN$DwsV7foFQAqQpx3UFbODLA==$0  TABLE      2024-09-30:11:17:35
EMP2               BIN$9dyGPwVPSg2LvVDuZTHzHA==$0  TABLE      2024-09-30:11:06:21
EMP2               BIN$fhw62iV8Sy2SapAZK+EPQQ==$0  TABLE      2024-09-30:10:57:21
SQL> FLASHBACK TABLE emp2 TO BEFORE DROP;
```

Flashback complete.

SQL> select * from emp2;

EMPID	ENAME	JOB	MGR	HIREDATE	SAL	COMM
7839	KING	PRESIDENT		17-NOV-81	5000	
10						
7788	BLAKE	MANAGER	7839	01-MAY-81	2850	
30						

25.Flashback the recently dropped table and restore it as EMP2_1.

SQL> FLASHBACK TABLE emp2 TO BEFORE DROP RENAME TO emp2_1;

Flashback complete.

```
SQL> FLASHBACK TABLE emp2 TO BEFORE DROP RENAME TO emp2_1;
```

```
Flashback complete.
```

```
SQL> select * from emp2_1;
```

EMPID	ENAME	JOB	MGR	HIREDATE	SAL	COMM
7839	KING	PRESIDENT		17-NOV-81	5000	
10						
7788	BLAKE	MANAGER	7839	01-MAY-81	2850	
30						
7799	Johnson	MANAGER	7342	01-JUN-87	2855	
30						

26.Display all tables owned by the current user.

```
SQL> SELECT table_name FROM user_tables;
```

```
TABLE_NAME
```

```
REGIONS
```

```
LOCATIONS
```

```
DEPARTMENTS
```

```
JOBS
```

```
EMPLOYEES
```

```
JOB_HISTORY
```

```
DEPT
```

```
EMP
```

```
LODGE_DETAILS
```

```
SKILL_DETAILS
```

```
EMP_DETAILS
```

```
TABLE_NAME
```

```
EMP_SKILL
```

```
DEPT11
```

```
EMP11
```

```
STUDENT
```

```
SALGRADE
```

```
DEPT_1
```

```
EMP_1
```

```
PRODUCTSTB
```

```
DEPT1
```

```
EMP2_1
```

COUNTRIES

TABLE_NAME

EMP_TEST

23 rows selected.